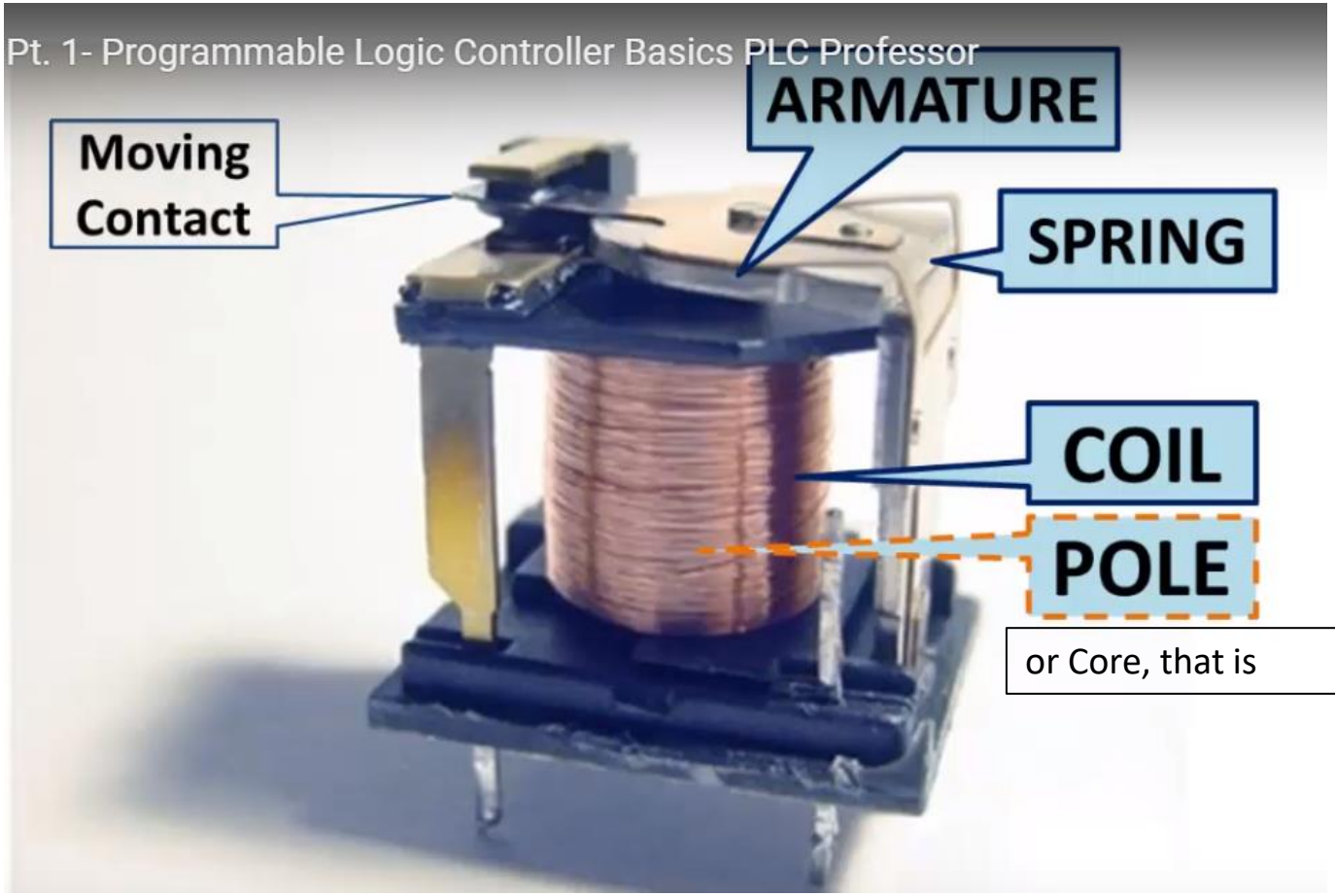
Relay Logic Ladder Diagram - 1

Those Old Times

Objectives

- Course Learning Objective #1: Students should recognize and understand the Relay Logic Programming and be able to compare it later with the Programmable Logic Control (PLC) of following modules.
- Specifically, at the end of this class, students should:
  - MLO #1.1: Describe the working principle and the physical parts of a relay.
  - MLO #1.2: Identify the poles and throws of a switch
  - MLO #1.3: Design relay logic to implement combinational logic (recognize open and closed relay contacts as digital signals)
  - MLO #1.4: Draw the ladder diagrams for combinational logic.
  - MLO #1.5: Recognize the basic elements (inputs and outputs) involved in relay ladder logic
  - MLO #1.6: Identify the parts of a ladder diagram (inputs and outputs) and recognize the diagramming symbols for common components (contacts, switches, etc.)
  - MLO #1.7: Implement the latch using ladder diagram (S-R flip-flop).

Relays-1


Pt. 1- Programmable Logic Controller Basics PLC Professor
ARMATURE
Moving Contact
SPRING
COIL
POLE
or Core, that is

Relays-2

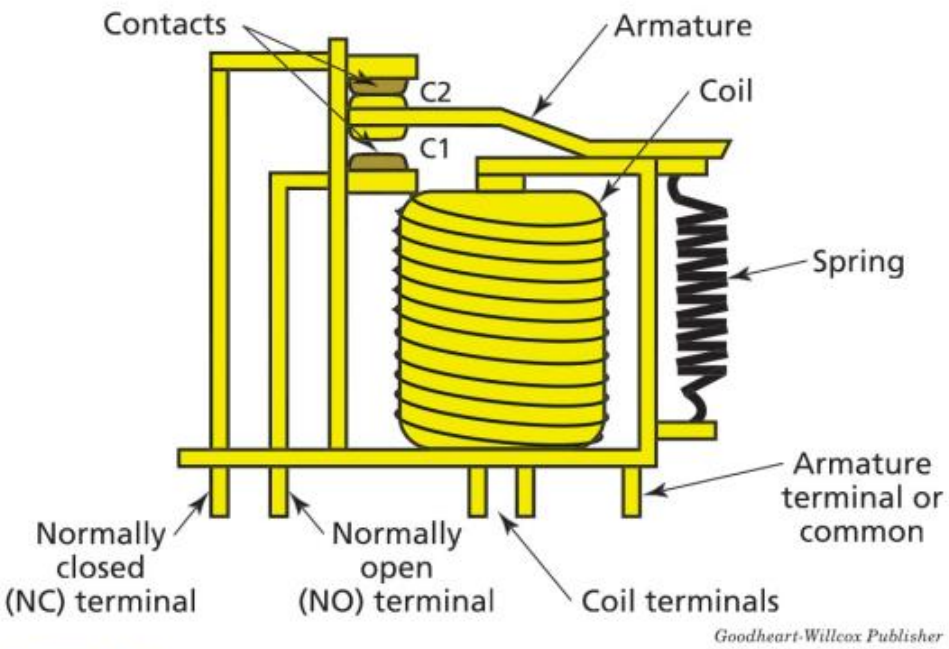
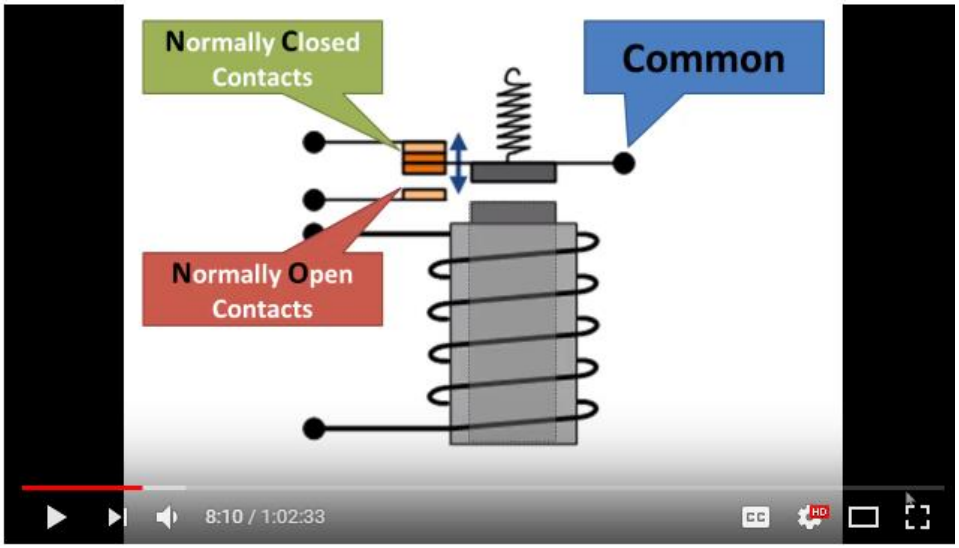


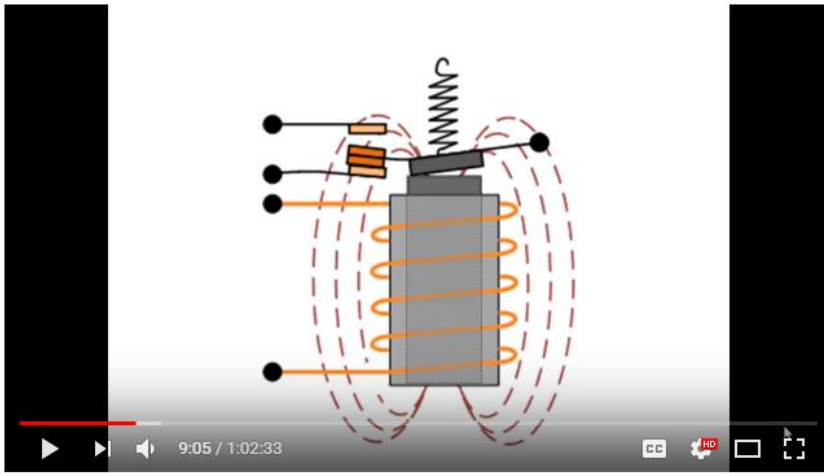
Figure 4-18. Construction of a relay.

Relays-3



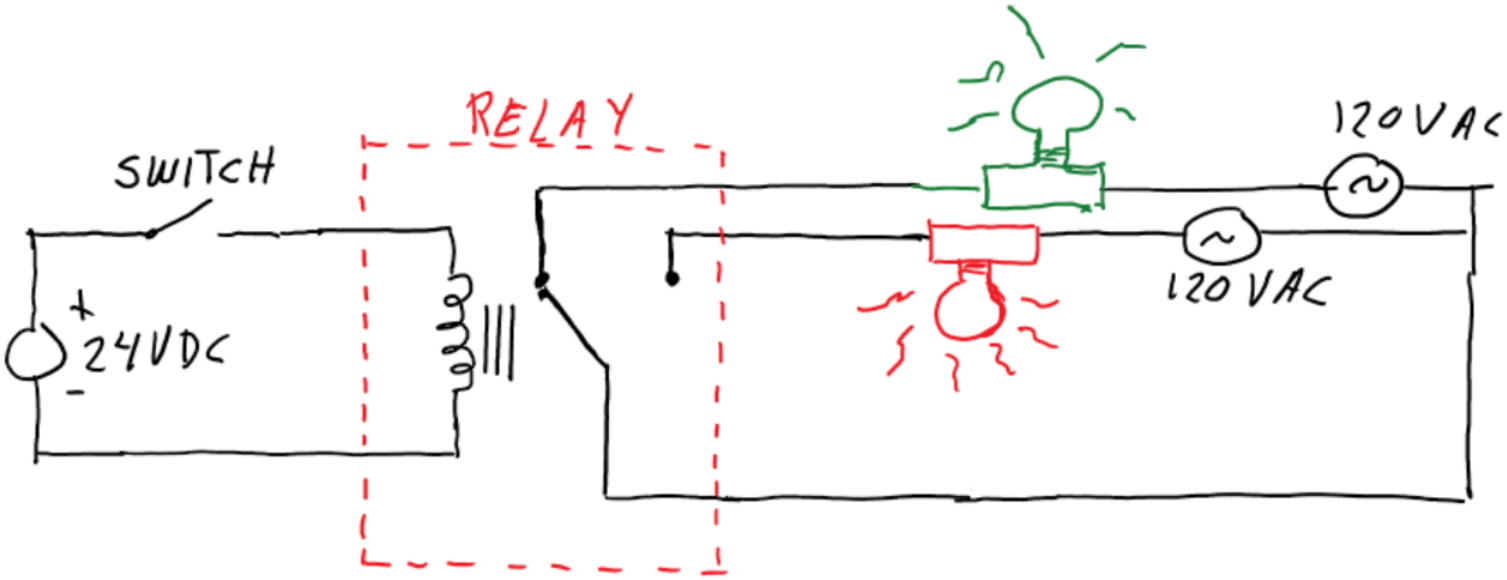
PLC Lecture 01 Pt. 1- Programmable Logic Controller Basics PLC Professor

A relay is a switch that can be manipulated by an electrical signal.



PLC Lecture 01 Pt. 1- Programmable Logic Controller Basics PLC Professor

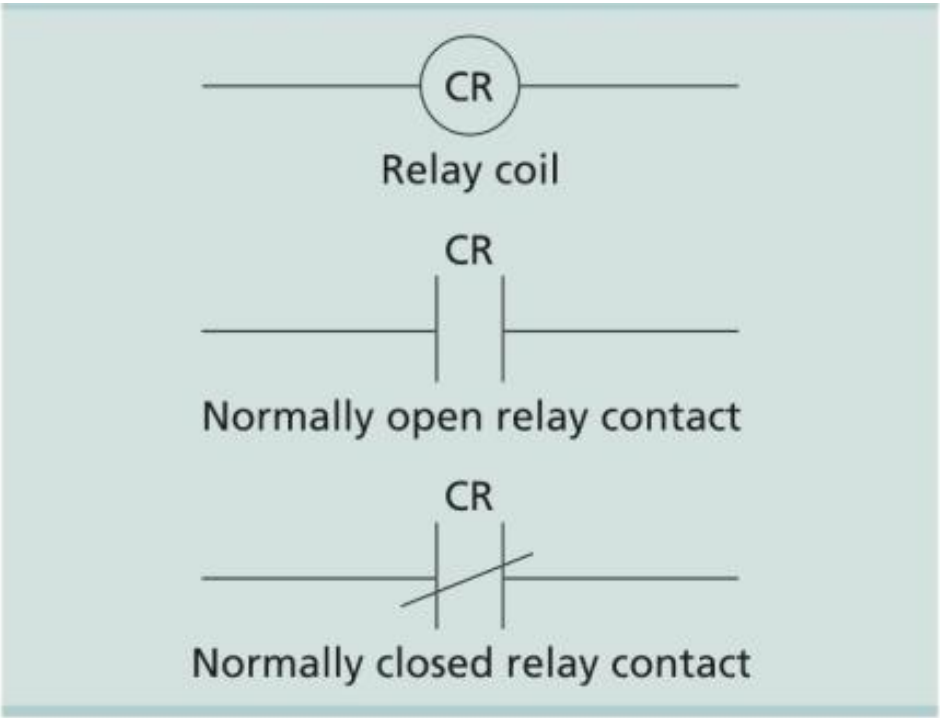
Relays-4



1. The switch is the input. It's a binary input: False (OFF, logical "0") when the switch is open and True (ON, logical "1") when the switch is closed.
2. When the input is "0" (switch open) the green light is ON, and the red light is OFF.
3. When the input is "1" (switch closed) the green light is OFF, and the right light is ON.
4. The input (switch) might be the result of a physical signal. For instance, when the engine temperature is normal the switch remains open; when the engine temperature is too high, the switch closes.

Relays-5

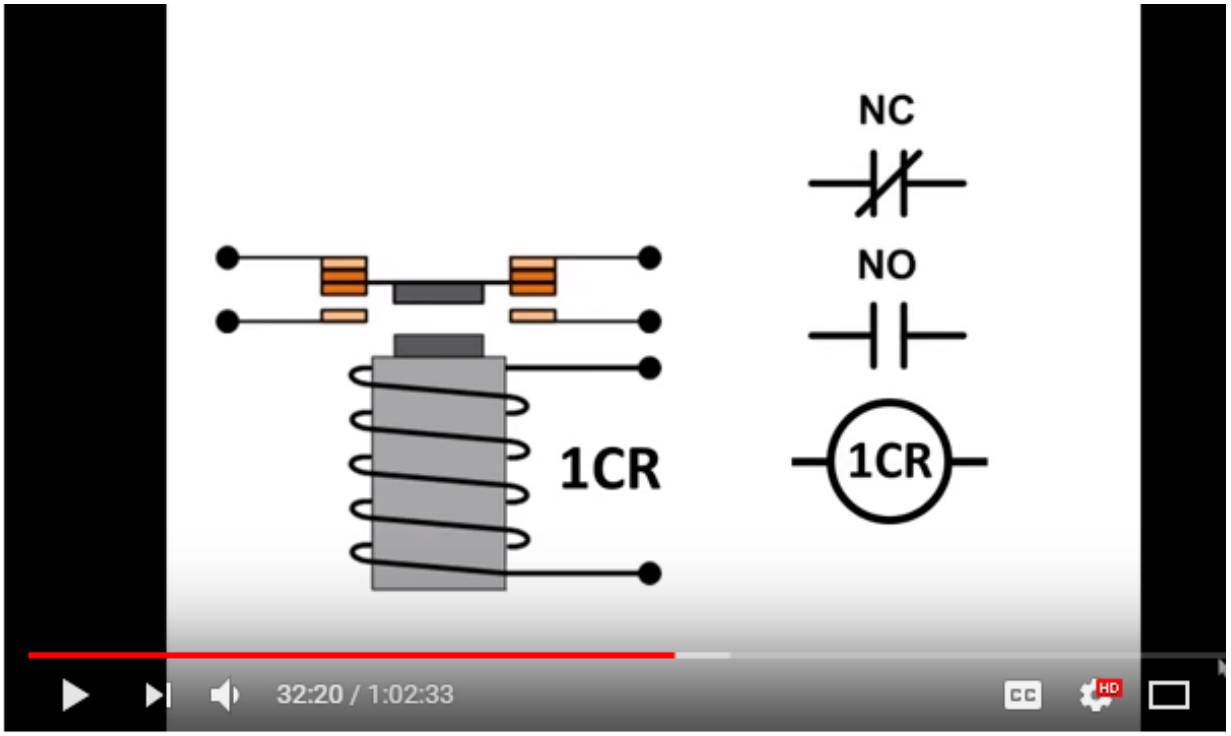
Relay Schematic



Goodheart-Willcox Publisher

Figure 4-19. Electrical symbols that represent the coil and contacts of a relay.

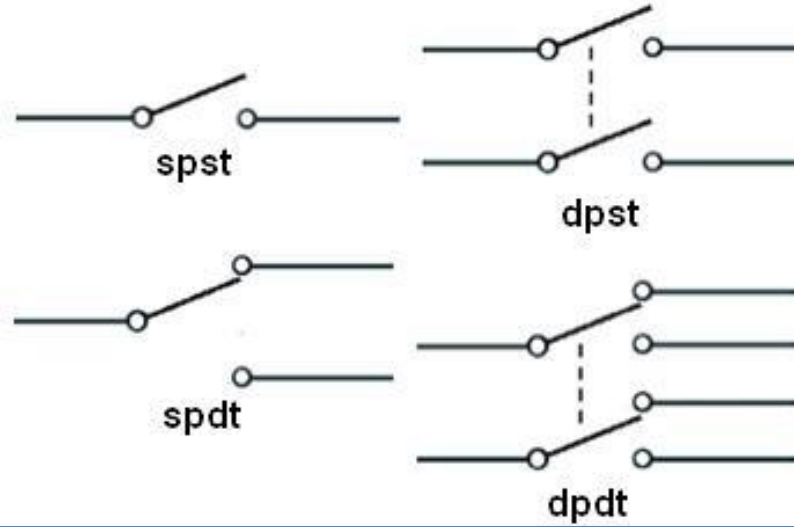
Relays-6



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Another way to identify schematically the contacts and coil of a relay

Relays (as Controllable Switches)-7



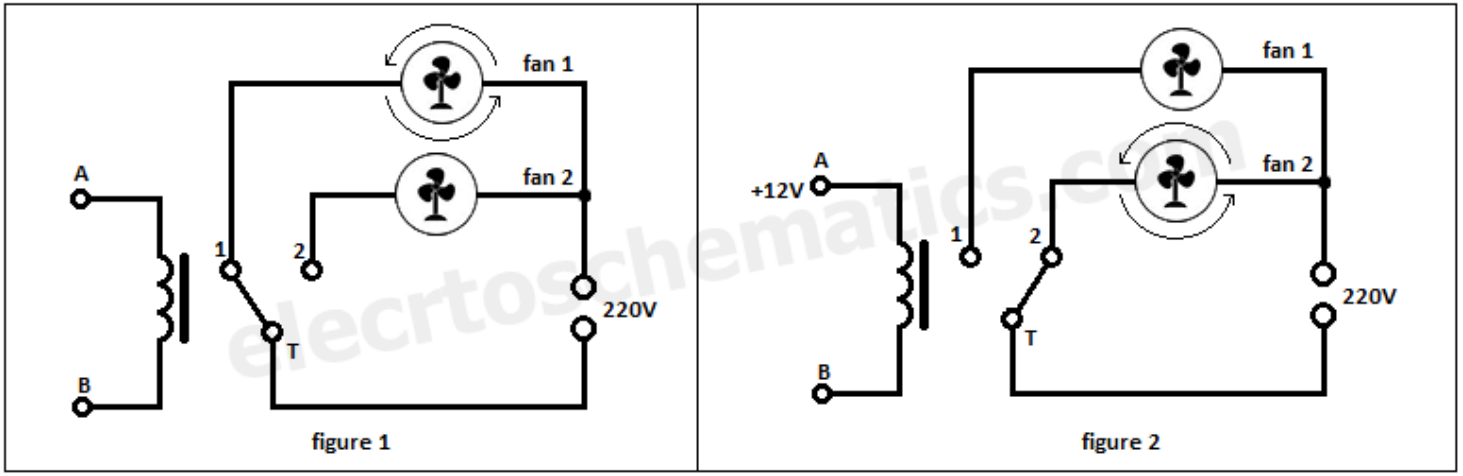
POLES: A switch pole refers to the number of switches the switch controls (a pole does not share a terminal with different circuits)

THROWS: Number of switch positions per pole (open position “in the air” does not count)

http://www.reuk.co.uk/wordpress/electronics/latching-relay-circuit/

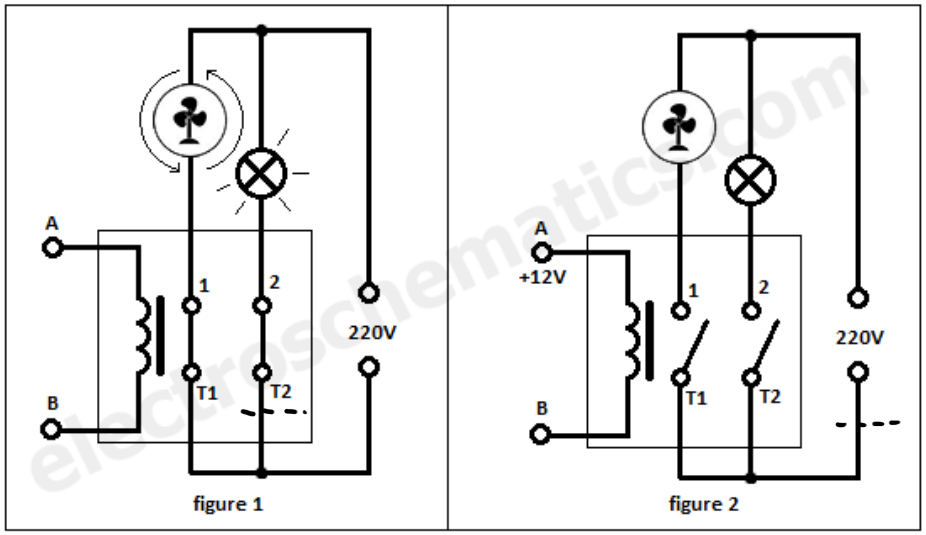
Relays-8

SPDT Relay



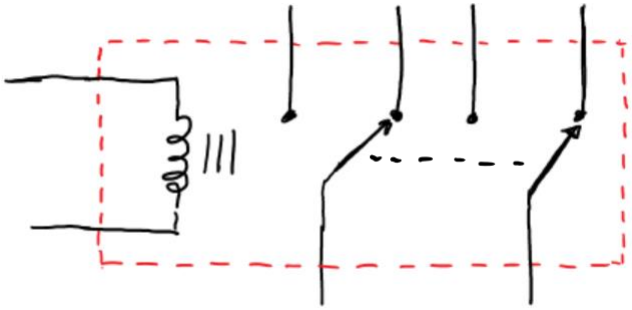
https://www.electroschematics.com/9605/dpst-switch-relay/

NC DPST Relay

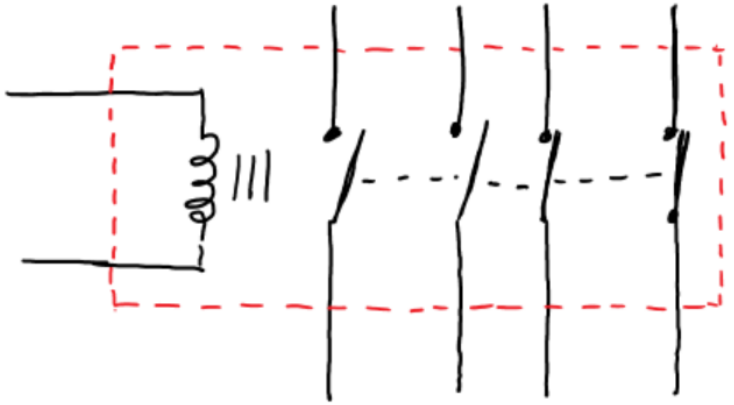


Relays-9

- Relays can have more than one NO contacts and more than one NC contacts to be able to control many circuits with one coil.



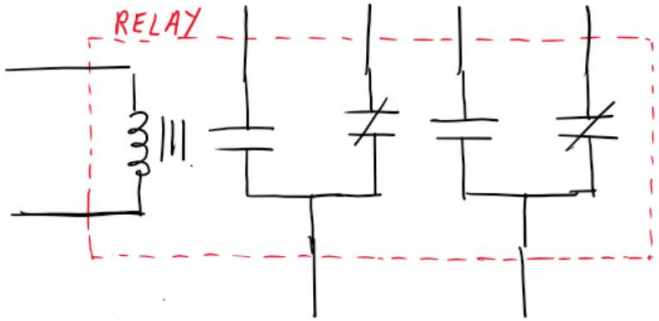
Two poles – Two throws



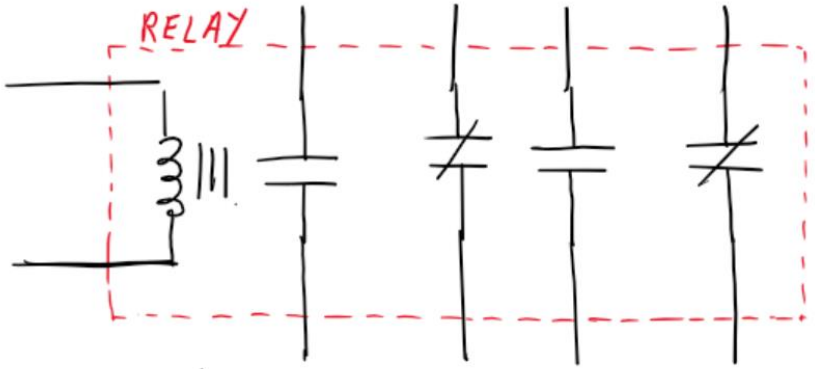
Four poles – One throw

Relays-10

- Instead of using switches we can use the relay contact representation in the schematic of the previous slide.

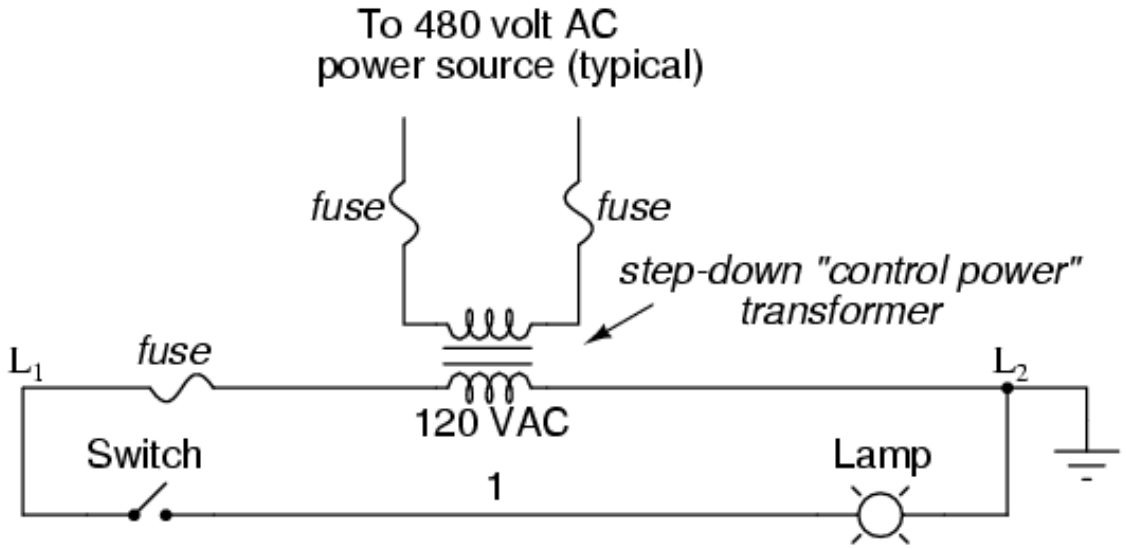


Two poles – Two throws



Four poles – One throw

Ladder Representation

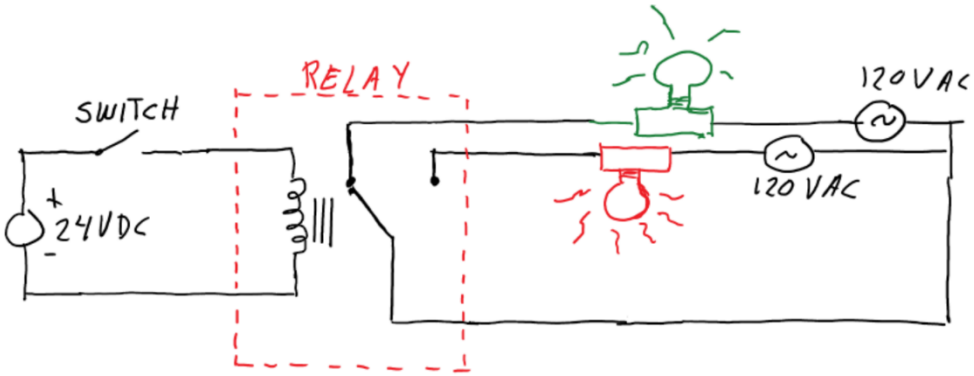


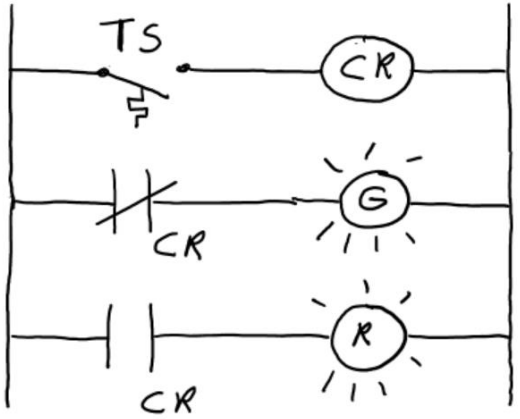
The lines L_1 and L_2 act as the sides of a ladder, and the horizontal line acts as one rung of the ladder. When the switch closes, the circuit is closed, and current goes through the rung turning the lamp on.

https://www.ibiblio.org/kuphaldt/electricCircuits/Digital/DIGI_6.html

Relay Ladder Representation

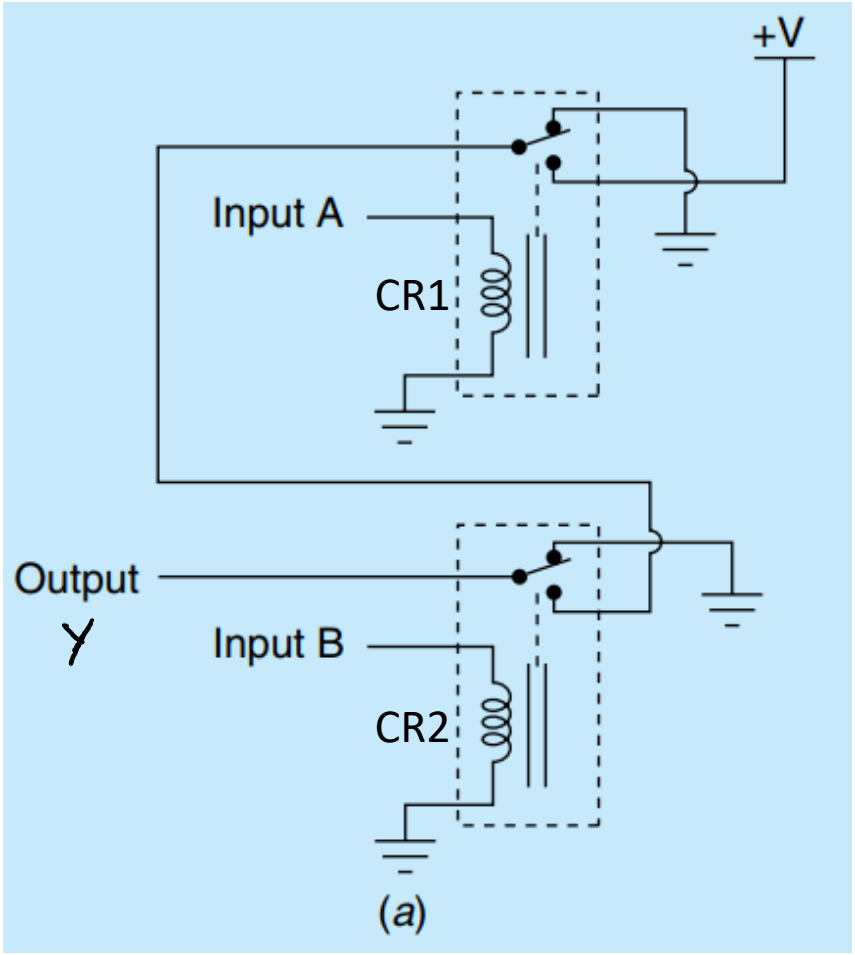
- Previous relay schematic and its ladder representation.





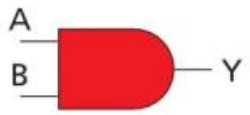
Ts: temperature switch. It closes when temperature is above certain threshold

AND Operation



1, ON, TRUE are equivalent
0,OFF, FALSE are equivalent

Boolean expression: **Y = A · B**

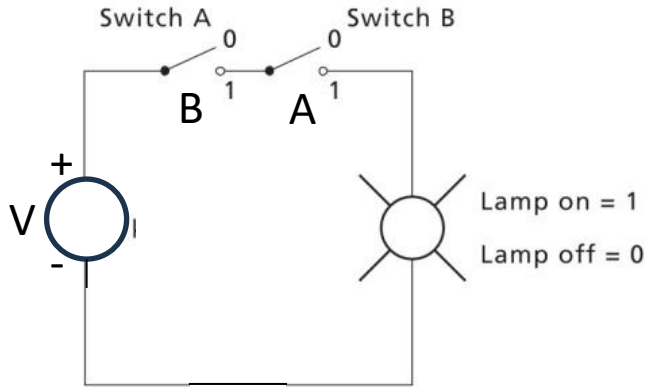


A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Two-input AND gate — Truth Table

Goodheart-Willcox Publisher

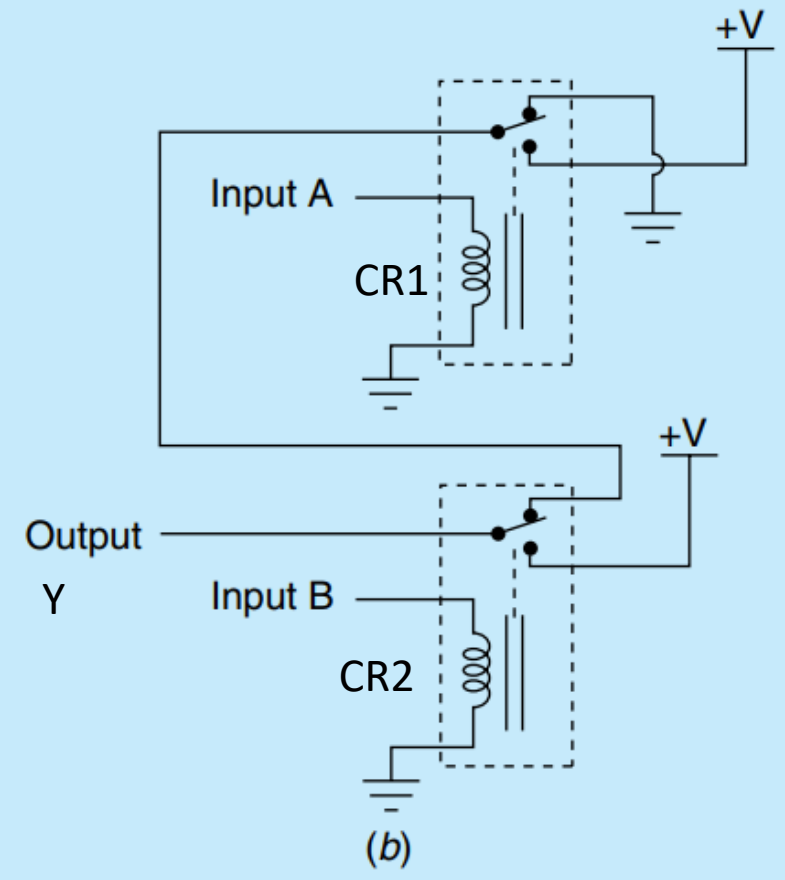
Figure 7-9. Boolean expression, gate symbol, and truth table for a two-input AND logic gate.



Goodheart-Willcox Publisher

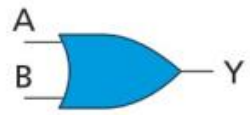
Figure 7-8. Electric circuit emulating an AND gate.

OR Operation



1, ON, TRUE are equivalent
0,OFF, FALSE are equivalent

Boolean expression: **Y = A + B**



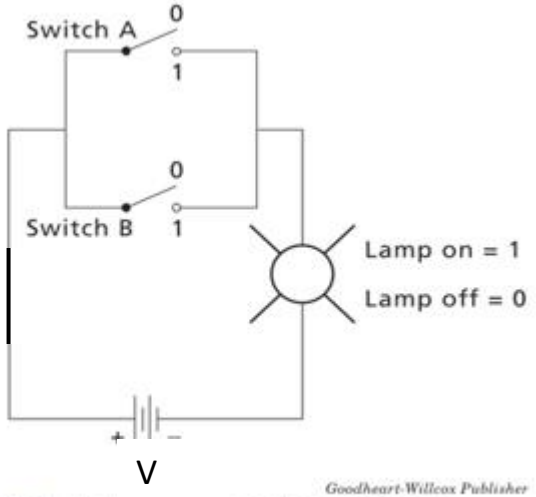
Two-input OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table

Goodheart-Willcox Publisher

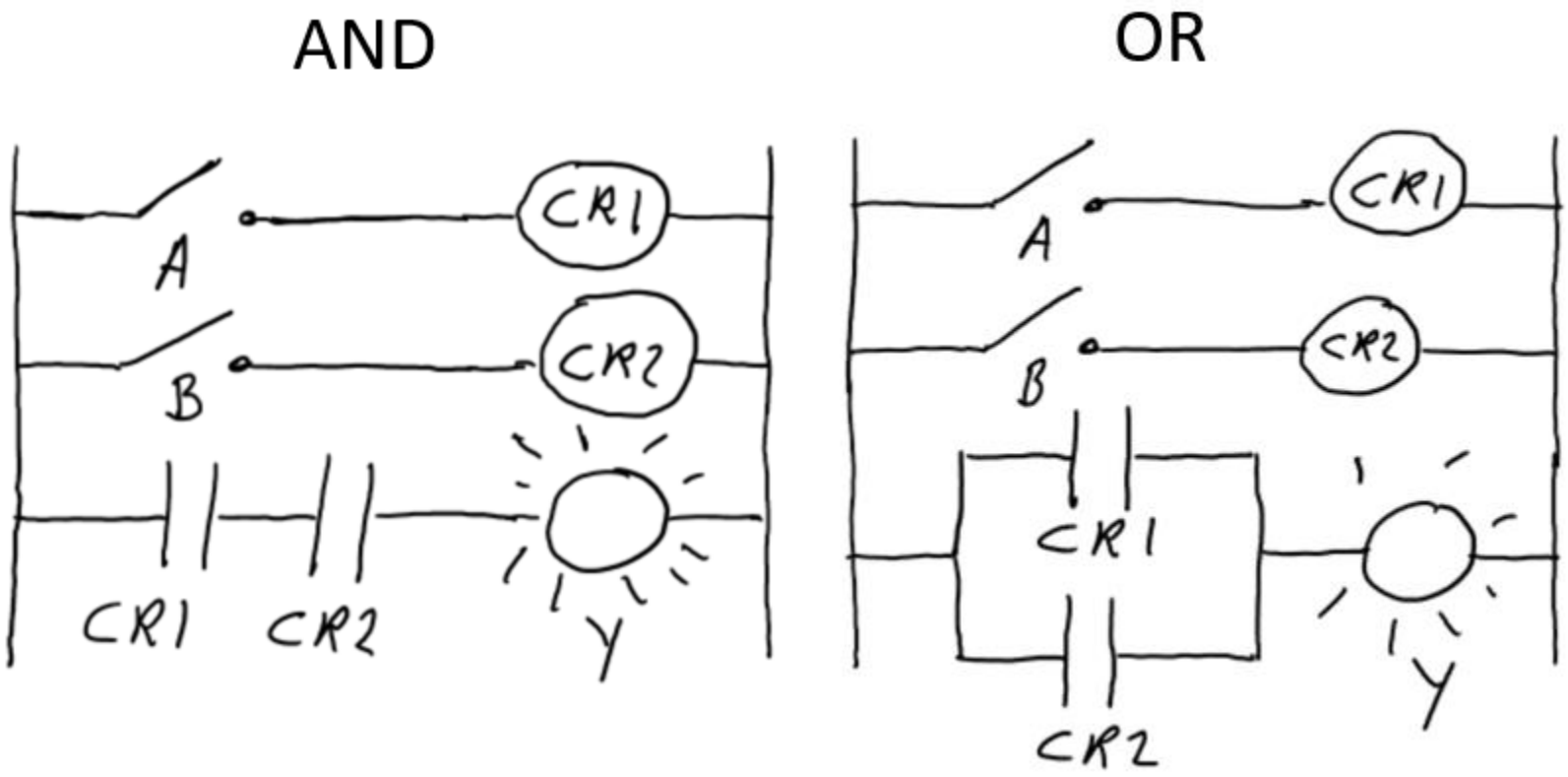
Figure 7-12. Boolean expression, gate symbol, and truth table for a two-input OR logic gate.



Goodheart-Willcox Publisher

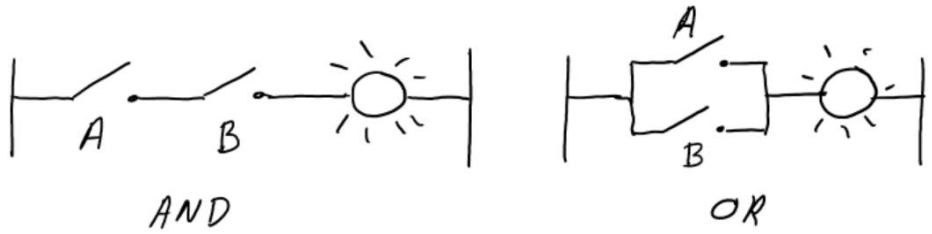
Figure 7-11. Electric circuit emulating an OR gate.

Relay Logic Ladder Diagram for AND and OR



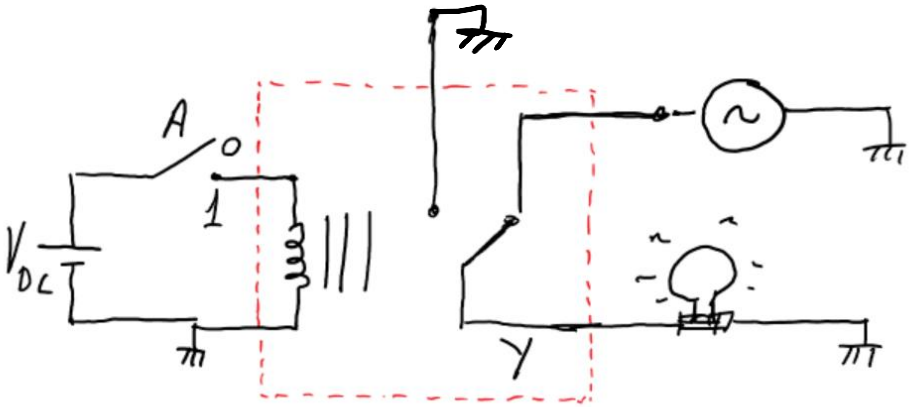
Relay Logic Ladder Diagram for AND and OR

- Notice that we could have used the switches to turn on the lamp directly (no relay used)

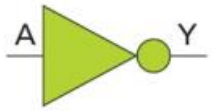


- But if we do, we couldn't use the switches on another rung (circuit) of the ladder.
- The relay contacts allow the use of the switches in more than one circuit (rung). Moreover, the relay NC contacts allow the use of the "NOT" operation of the switch as indicated in the next slide.

The NOT operation



Boolean expression: $Y = \overline{A}$



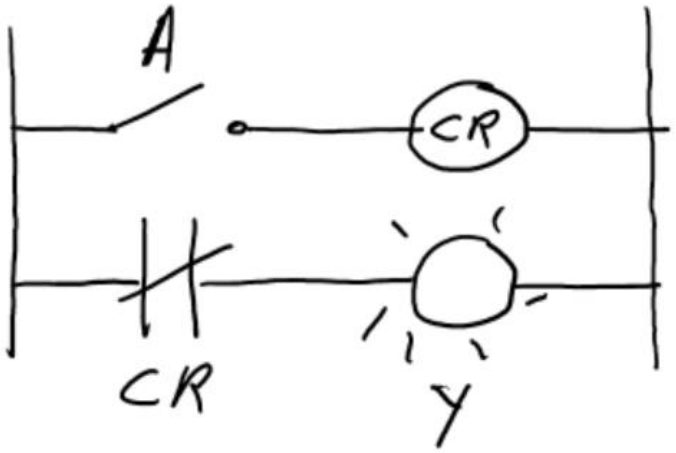
NOT gate symbol

A	Y
0	1
1	0

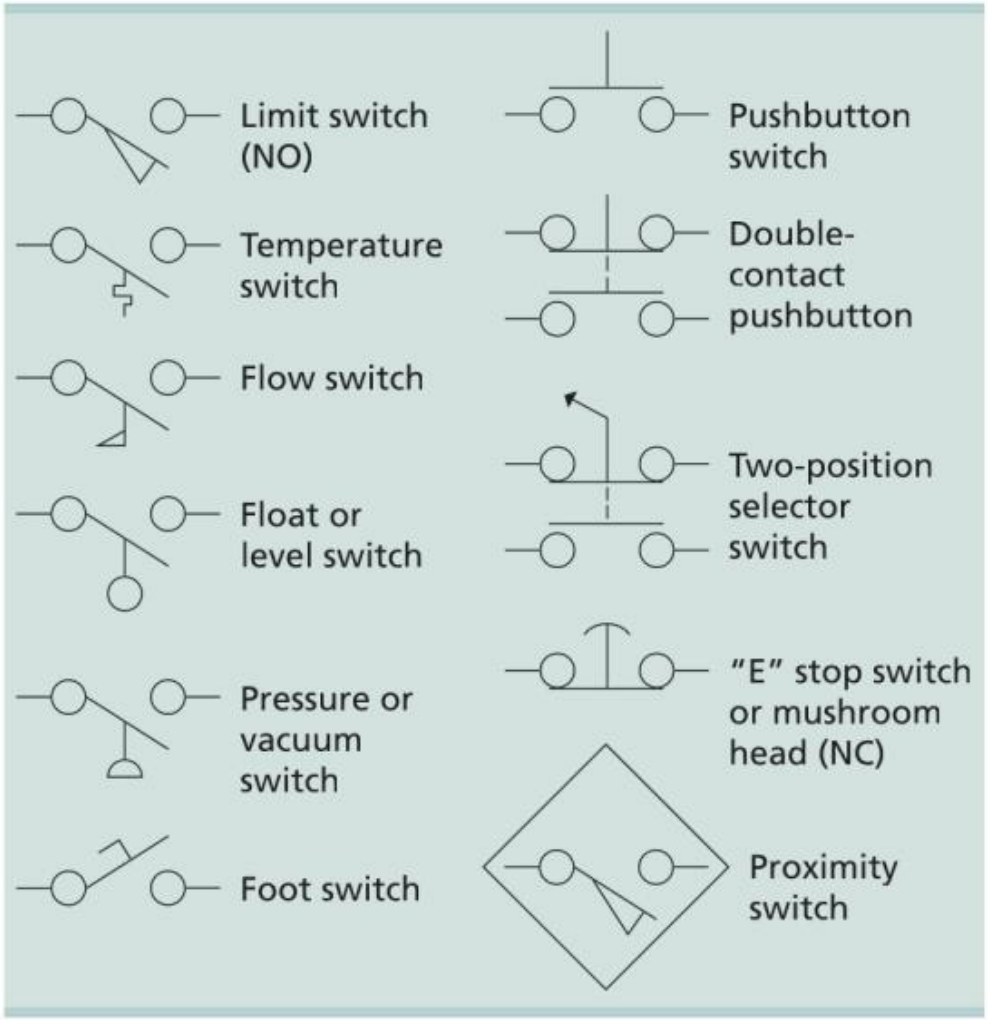
Truth Table

Goodheart-Willcox Publisher

Figure 7-6. Boolean expression, gate symbol, and truth table for a NOT logic gate.



Some Switch (Inputs) Symbols used in Relay Logic Diagram (RLD)



Goodheart-Willcox Publisher

Figure 4-4. Electrical symbols for some common switches.

Some Output Elements used in Relay Logic Diagrams

Indicator lamp



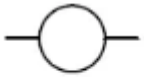
Heating element



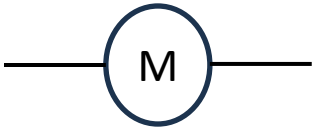
Solenoid coil



Relay coil



Motor Starter Coil



Example 1

- Implement the following binary equation as a an RLD knowing that A and B are NO limit switches (inputs) and Z (output) is a motor starter.

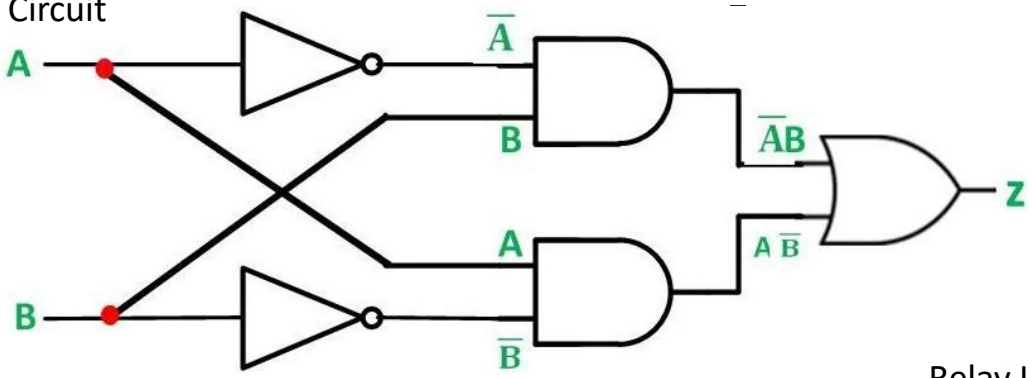
$$\mathrm{Z} = A\bar{B} + \bar{A}B$$

- Always keep in mind that inputs are on the left and output are on the right.

Example 1 (EXOR)

Logical Circuit

Boolean expression: $Z = A \oplus B = A \bullet \overline{B} + \overline{A} \bullet B$



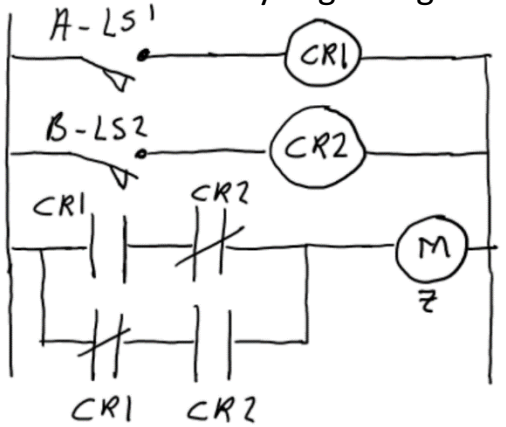
Truth Table

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

Expanded Truth Table

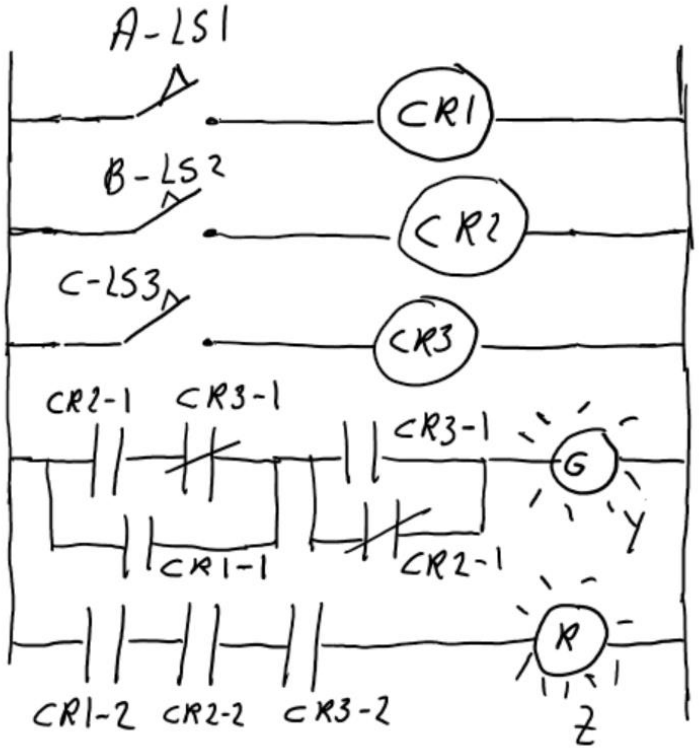
A	$\overline{A}$	B	$\overline{B}$	$A\overline{B}$	$\overline{A}B$	Z
0	1	0	1	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	0	0	0	0

Relay Logic diagram



Example 2

- Write the logic equations (notice that there are two outputs and three inputs) resulting from the following ladder diagram



Example 2

- By inspection, we obtain

$$Y = (B\bar{C} + A)(C + \bar{B})$$
$$Z = ABC$$

- Logic equations can be simplified but we will not try it here. We found the equations by inspection.
- Using the equations without simplification, we notice that NO contacts CR1, CR2 and CR3 are used twice.

Example 2

- Physically, a pair of contacts can be used only once. This means that the relays CR1, CR2 and CR3 have to have at least two pairs of open contacts (2 poles)
- The RLD shows the second pair of contacts as CR1-2, CR2-2 and CR3-2. They have the subindex "-2." It's OK to omit such index because it's understood that every individual circuit needs different contacts.
- We will deal with the equation simplification in class.

Latch Circuit and Motor Starter

- The diagrams show a three-phase motor starter. It needs three pair of contacts provided by the motor starter relay.
- OL stands for Over-Load. When the motor is overloaded, requiring too much current to operate, the NC overload contacts will open (they act as a fuse)
- When the start push-button is pressed and released, the motor turns on. If the stop button is, then, pressed and released, the motor will stop.

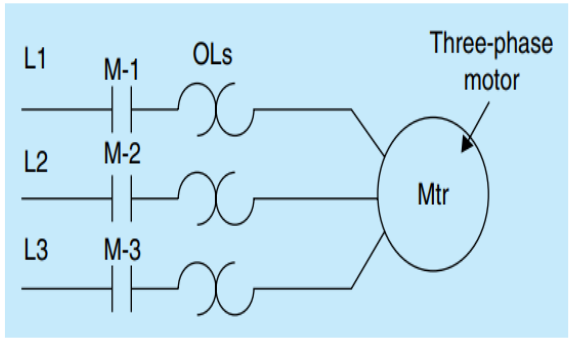


Figure 1.13 High-power motor circuit.

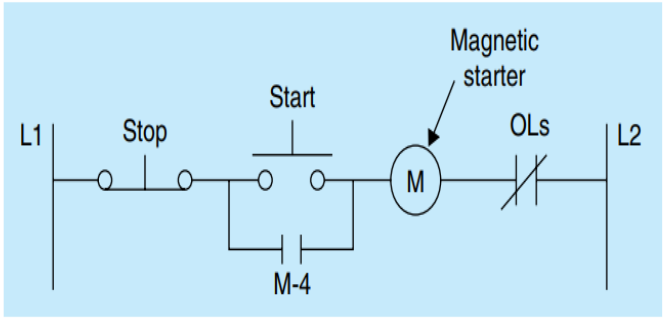
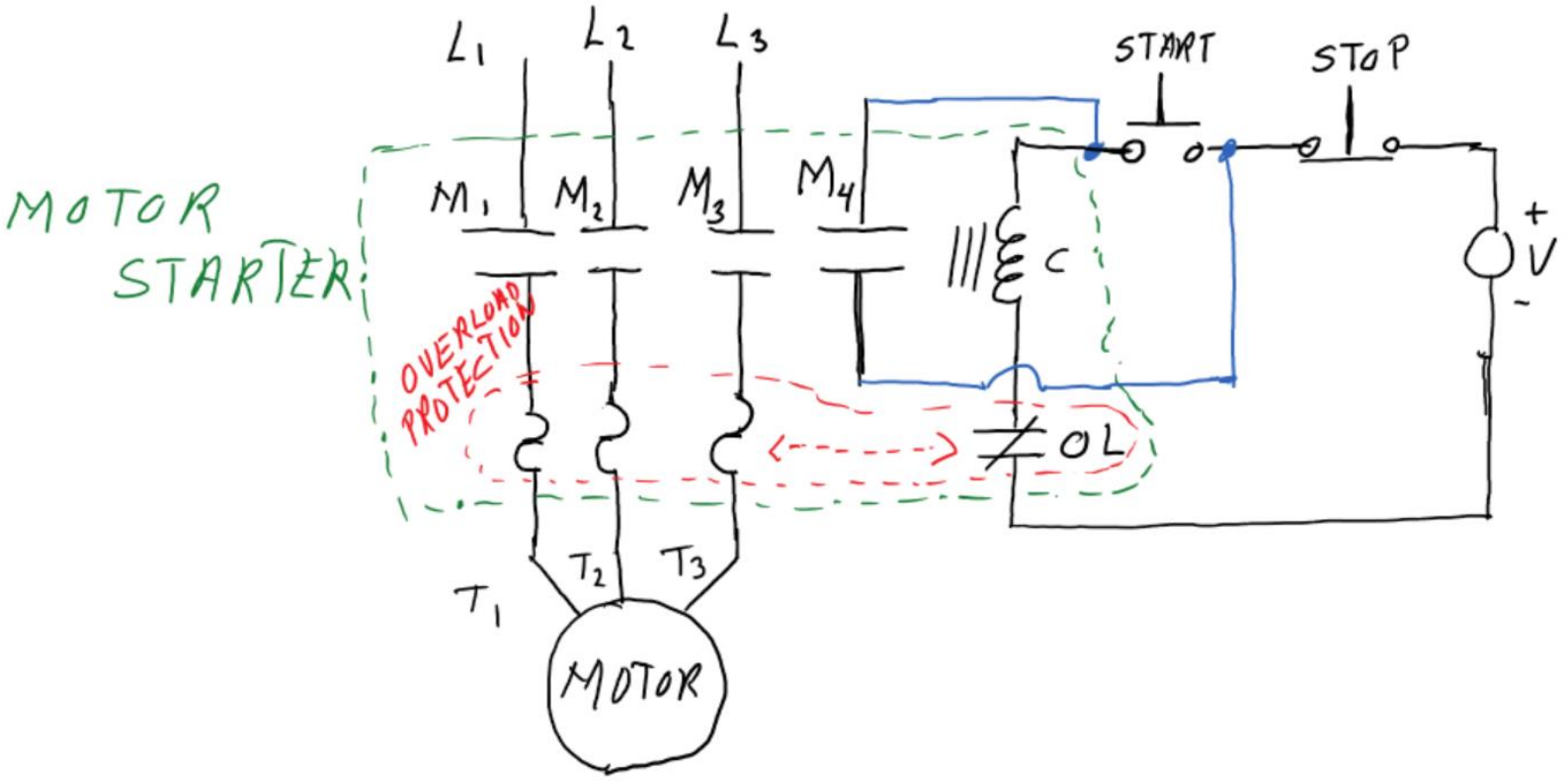


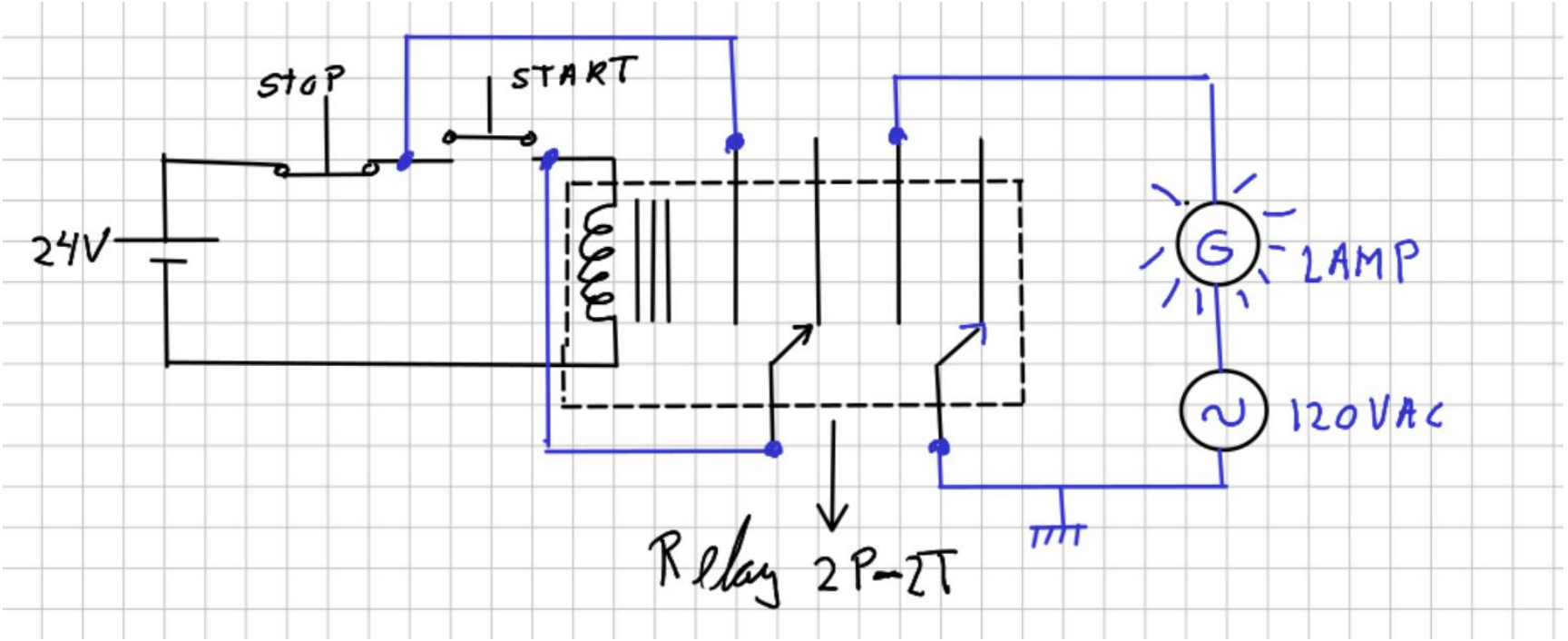
Figure 1.14 Low-power motor starter circuit.

Latch Circuit and Motor Starter



Latch Circuit for a Lamp

- We need a relay with two poles and one throw. (However, the fig. shows the most typical two throws relay. Notice that the other throw is not used.)
- One pole handles the START switch, the other the lamp circuit.
- Remember, one pair of contacts can be used only for one circuit.



Relay comes from relay race



Relays used for telegraph

